

LC: 118 Pascal's Triangle

Simulation

Observation:-

- See the triangle as right angle triangle
- Row length = row number
- Every row start & ends with 1

Logic:- ① $\text{ans}[i].\text{resize}(i+1, 1)$

② Compute middle values using prev rows $\text{ans}[i][j] = \text{ans}[i-1][j-1] + \text{ans}[i-1][j]$

③ Repeat for every row

Trigger:- See as right angle triangle

Complexity $\rightarrow$ TC: $O(n^2)$ SC: $O(n^2)$

LC: 73 Set Matrix ZerosIn Place
Matrix Marking

Naive:- Use 2 arrays for row & col marking, but SC: $O(m+n)$

Optimal / Best

Observation:- ① We can use first row & first col as marker arrays.

② Need one extra variable (col0) cuz $\text{matrix}[0][0]$ cannot represent both row 0 & col 0

Logic:- ① Traverse matrix, if $\text{matrix}[i][j] == 0 \rightarrow$ mark it $\rightarrow \text{C111} \rightarrow$ end not markers.

② Now use markers to set internal cells 0

③ If $\text{matrix}[0][0] == 0 \rightarrow$ whole marker for col is 0

④ If $\text{col0} == 0 \rightarrow$ whole marker for row is 0

Trigger:- Use 0th row for cols & 0th col for row marking and keep col0 separated

Complexity:- TC:- $O(nm)$ SC:- $O(1)$

LC 31: Next PermutationGreedy +
Array ManipulationObservation:- We want slightly bigger ele than curr.

• Suppose we have 146532

• To make slightly bigger we modify right side not left

• 32 can't be changed, 532 x, 6532 x, 146532 ✓

• We want first bigger ele than 4 which is 5, swap them 56432

→ Now we want this part as smallest, so sort it or reverse it

Logic:- ① Traverse from right & find pivot

② Find smallest ele greater than pivot

③ Swap pivot, that ele

④ Reverse it.

Trigger:- Find Pivot → swap with next larger ele → reverse after pivot.Complexity:- Tc: $O(n)$, Sc: $O(1)$